Assignment 2 Solution

1. Point Estimation (15 pts)

The Poisson distribution is a useful discrete distribution which can be used to model the number of occurrences of something per unit time. For example, in networking, packet arrival density is often modeled with the Poisson distribution. If X is Poisson distributed, i.e., $X \sim Poisson(\lambda)$, its probability mass function takes the following form:

$$P(X|\lambda) = \frac{\lambda^X e^{-\lambda}}{X!}.$$

It can be shown that $\mathbb{E}(X) = \lambda$. Assume now we have n i.i.d. data points from $Poisson(\lambda)$: $\mathcal{D} = \{X_1, \cdots, X_n\}$. (For the purpose of this problem, you can only use the knowledge about the Poisson and Gamma distributions provided in this problem.)

(a) Show that the sample mean $\hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is the maximum likelihood estimate (MLE) of λ and it is unbiased ($\mathbb{E}\hat{\lambda} = \lambda$).

(b) Now let's be Bayesian and put a prior distribution over λ. Assuming that λ follows a Gamma distribution with the parameters (α, β), its probability density function:

$$p(\lambda|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda},$$

Where $\Gamma(\alpha) = (\alpha-1)!$ (here we assume α is a positive integer). Compute the posterior distribution over λ.

(c) Derive an analytic expression for the maximum a posterior (MAP) of λ under $Gamma(\alpha, \beta)$ prior.

★ **SOLUTION:**

(a) Write down the log-likelihood

$$\ln P(\mathcal{D}|\lambda) = \ln e^{-n\lambda}\prod_{i=1}^{n}\frac{\lambda^{X_i}}{X_i!} = -n\lambda + \sum_{i=1}^{n}\{X_i \ln\lambda - \ln(X_i!)\}.$$

The MLE $\hat{\lambda} = \arg\max_\lambda P(\mathcal{D}|\lambda) = \arg\max_\lambda \ln P(\mathcal{D}|\lambda)$, which can be obtained by setting the gradient of $\ln P(\mathcal{D}|\lambda)$ with respect to λ to 0. More specifically:

$$\frac{d}{d\lambda}\ln P(\mathcal{D}|\lambda) = -n + \frac{1}{\lambda}\sum_{i=1}^{n} X_i = 0 \Longrightarrow \hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Since $X_1, \ldots, X_n$ are i.i.d. from $Poisson(\lambda)$, for any X_i, $\mathbb{E}(X_i) = \lambda$. $\hat{\lambda}$ is unbiased because:

$$\mathbb{E}(\hat{\lambda}) = \mathbb{E}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(X_i) = \frac{1}{n}\sum_{i=1}^{n}\lambda = \lambda$$

(b)

$$\begin{aligned} p(\lambda|\mathcal{D}) &\propto P(\mathcal{D}|\lambda)p(\lambda) \\ &= e^{-n\lambda}\left(\prod_{i=1}^{n}\frac{\lambda^{X_i}}{X_i!}\right)\frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda} \\ &\propto \lambda^{\sum_{i=1}^{n} X_i + \alpha - 1}e^{-n\lambda - \beta\lambda} \end{aligned}$$

Therefore, the posterior distribution $p(\lambda|\mathcal{D}) \sim Gamma(\sum_{i=1}^{n} X_i + \alpha, n + \beta)$

2. Source of Error: Part 1 (15 pts)

Suppose that we are given an independent and identically distributed sample of n points $\{y_i\}$ where each point $y_i \sim \mathcal{N}(\mu, 1)$ is distributed according to a normal distribution with mean μ and variance 1. You are going to analyze different estimators of the mean μ.

(a) Suppose that we use the estimator $\hat{\mu} = 1$ for the mean of the sample, ignoring the observed data when making our estimate. Give the bias and variance of this estimator $\hat{\mu}$. Explainin a sentence whether this is a good estimator in general, and give an example of when this is a good estimator.

(b) Now suppose that we use $\hat{\mu} = y_1$ as an estimator of the mean. That is, we use the first data point in our sample to estimate the mean of the sample. Give the bias and variance of thisestimator $\hat{\mu}$. Explain in a sentence or two whether this is a good estimator or not.

(c) In the class you have seen the relationship between the MLE estimator and the least squares problem. Sometimes it is useful to use the following estimate

$$\hat{\mu} = \arg\min_{\mu} \sum_{i=1}^{n} (y_i - \mu)^2 + \lambda\mu^2$$

For the mean, where the parameter $\lambda > 0$ is a known number. The estimator $\hat{\mu}$ is biased, but has lower variance than the sample mean $\bar{\mu} = n^{-1}\sum_i y_i$ which is an unbiased estimator for μ. Give the bias and variance of the estimator $\hat{\mu}$.

★ **SOLUTION:** The bias of an estimator is defined as $E[\hat{\mu}] - \mu$. Since we have that $E[\hat{\mu}] = 1$, the bias is $1 - \mu$.

The variance of an estimator is defined as $\mathrm{Var}(\hat{\mu}) = E[(\hat{\mu} - E[\hat{\mu}])^2]$. Therefore, plugging in $\hat{\mu} = 1$, we have that $\mathrm{Var}(\hat{\mu}) = 0$.

This is not a good estimator, since the bias is large when the true value of μ is not 1. Usually we don't have any information about the true value of μ, so it is unreasonable to assume it is equal to 1.

★ **SOLUTION:** We have $E[\hat{\mu}] = \mu$. Therefore, the bias is 0. This is an unbiased estimator.

The variance of this estimator is $\mathrm{Var}(\hat{\mu}) = \mathrm{Var}(y_1) = 1$.

Although this estimator is unbiased, this is not a good estimator since its variability does not decrease with the sample size. For example, variance of the sample mean decreases as $1/n$.

★ **SOLUTION:** First, we need to find a closed form for $\hat{\mu}$. Taking the derivative of the objective with respect to μ and setting it equal to zero, we obtain that

$$-2\sum_{i=1}^{n}(y_i - \mu) + 2\lambda\mu = 0.$$

Solving for μ we have that

$$\hat{\mu} = \frac{1}{n+\lambda}\sum_i y_i = \frac{n}{n+\lambda}\bar{\mu}.$$

For this estimator

$$E[\hat{\mu}] = \frac{1}{n+\lambda}E[\sum_i y_i] = \frac{n}{n+\lambda}\mu.$$

Therefore

$$\text{bias} = \frac{-\lambda\mu}{n+\lambda}.$$

The bias decreases to 0 with the sample size.
For variance, we have that

$$\text{Var}(\hat{\mu}) = \text{Var}(\frac{1}{n+\lambda}\sum_i y_i) = \frac{1}{(n+\lambda)^2}\sum_i \text{Var}(y_i) = \frac{n}{(n+\lambda)^2}.$$

Comparing to the variance of the sample mean, which is $1/n$, we observe that the variance of $\hat{\mu}$ is smaller.

3. Source of Error: Part 2 (15 pts)

In class we discussed the fact that machine learning algorithms for function approximation are also a kind of estimator (of the unknown target function), and that errors in function approximation arise from three sources: bias, variance, and unavoidable error. In this part of the question you are going to analyze error when training Bayesian classifiers.

Suppose that Y is boolean, X is real valued, $P(Y=1) = 1/2$ and that the class conditional distributions $P(X|Y)$ are uniform distributions with $p(X|Y=1) = uniform[1,4]$ and $p(X|Y=0) = uniform[-4,-1]$. (we use $uniform[a,b]$ to denote a uniform probability distribution between a and b, with zero probability outside the interval $[a,b]$).

(a) Plot the two class conditional probability distributions $p(X|Y=0)$ and $p(X|Y=1)$.

(b) What is the error of the optimal classifier? Note that the optimal classifier knows $P(Y=1)$, $p(X|Y=0)$ and $p(X|Y=1)$ perfectly, and applies Bayes rule to classify new examples. Recall that the error of a classifier is the probability that it will misclassify a new x drawn at random from $p(X)$. The error of this optimal Bayes classifier is the unavoidable error for this learning task.

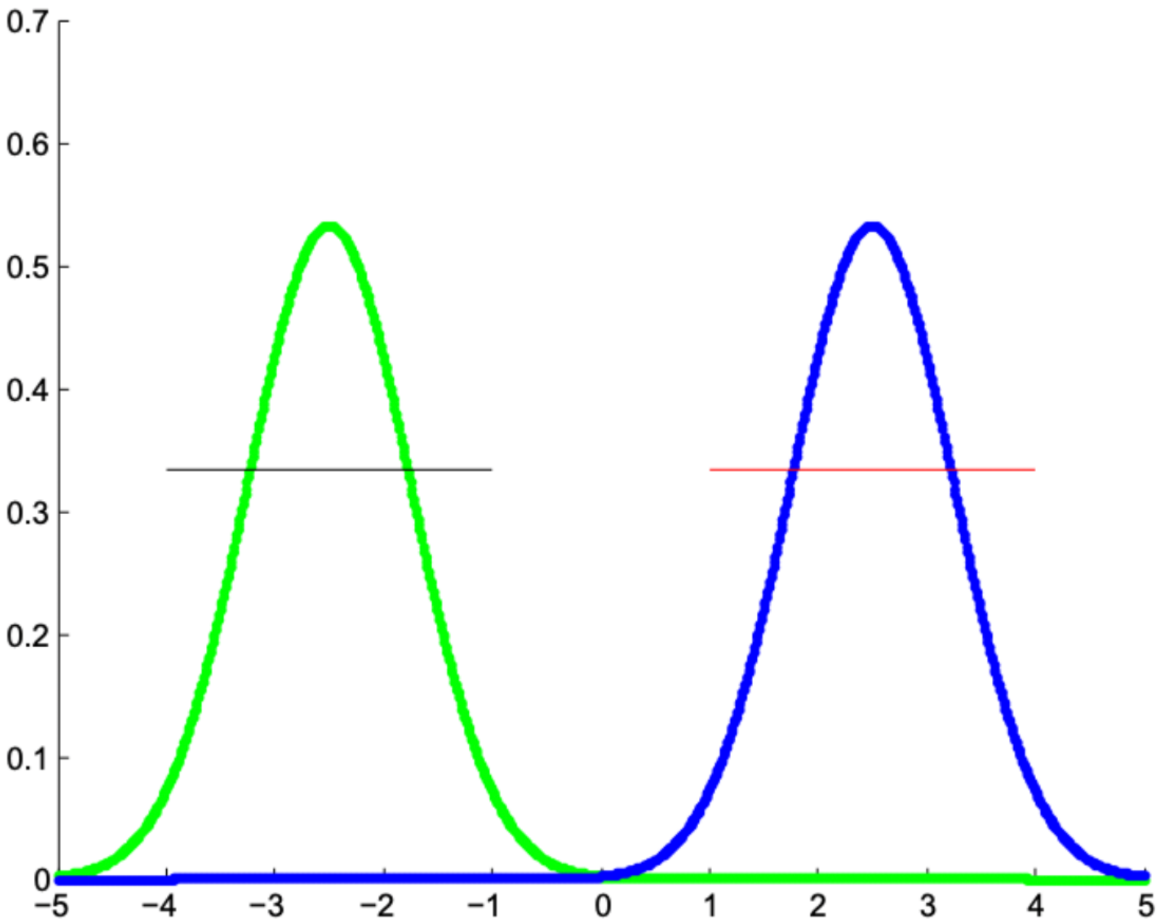
(c) Suppose instead that $P(Y=1) = 1/2$ and that the class conditional distributions are uniform distribution with $p(X|Y=1) = uniform[0,4]$ and $p(X|Y=0) = uniform[-3,1]$. What isthe unavoidable error in this case? Justify your answer.

(d) Consider again the learning task from part (a) above. Suppose we train a Gaussian Naive Bayes (GNB) classifier using n training examples for this task, where $n \to \infty$. Of course our classifier will now (incorrectly) model $p(X|Y)$ as a Gaussian distribution, so it will be biased: it cannot even represent the correct form of $p(X|Y)$ or $P(Y|X)$.

Draw again the plot you created in part (a), and add to it a sketch of the learned/estimated class conditional probability distributions the classifier will derive from the infinite training data. Write down an expression for the error of the GNB. (hint: your expression will involve integrals - please don't bother solving them).

(e) So far we have assumed infinite training data, so the only two sources of error are bias and unavoidable error. Explain in one sentences how your answer to part (d) above would change if the number of training examples was finite. Will the error increase or decrease? Which of the three possible sources of error would be present in this situation?

Solution:



The black line denotes $P[X|Y=0]$ and the red line denotes $P[X|Y=1]$. Green and blue lines denote $\hat{P}[X|Y=0]$ and $\hat{P}[X|Y=1]$, approximations using the Gaussian distribution in part d.

★ **SOLUTION:** The error of this classifier is equal to 0. We observe that supports of $p(X|Y=0)$ and $p(X|Y=1)$ do not overlap. Therefore, we can perfectly classify a new example, just by looking whether it is in the interval $[-4,-1]$ or in the interval $[1,4]$.

★ **SOLUTION:** In this case we will make an error if $x \in [0,1]$. An error of a perfect classifier in when $x \in [0,1]$ is equal to $1/2$. Therefore,

$$\begin{aligned}
P[\text{error}] &= P[x \in [0,1]] * P[\text{error}|x \in [0,1]] \\
&= (P[x \in [0,1]|y=0]P[y=0] + P[x \in [0,1]|y=1]P[y=1]) * P[\text{error}|x \in [0,1]] \\
&= (\frac{1}{4}\frac{1}{2} + \frac{1}{4}\frac{1}{2}) * \frac{1}{2} \\
&= \frac{1}{8}
\end{aligned}$$

★ **SOLUTION:** Given that we have infinite amount of data, we can compute $E[X|Y=0]=-2.5$ and $\mathrm{Var}[X|Y=0]=3/4$ (using formula for variance of the uniform distribution) and $E[X|Y=1]=2.5$ and $\mathrm{Var}[X|Y=1]=3/4$. Since we are approximating $p(X|Y)$ with the Normal distribution, we have that $\hat{p}(X|Y=0)=N(-2.5,0.75)$ and $\hat{p}(X|Y=1)=N(2.5,0.75)$.

Using this, we have that for $x<0$, $\hat{p}(X|Y=0)>\hat{p}(X|Y=1)$ and for $x>0$, we have that $\hat{p}(X|Y=0)<\hat{p}(X|Y=1)$. Therefore, the classifier will make no error when classifying new points. This example illustrates that even with incorrect model assumption, we can perform well when classifying examples.

★ **SOLUTION:** Given finite amount of data we will not perfectly learn mean and variance of $p(X|Y)$. Therefore, the error of the classifier will increace given finite amount of data. We would have bias and error in this situation.

4. Gaussian (Naïve) Bayes and Logistic Regression (15 pts)

Recall that a generative classifier estimates $P(\boldsymbol{X},Y)=P(Y)P(\boldsymbol{X}|Y)$, while a discriminative classifier directly estimates $P(Y|\boldsymbol{X})$. (Note that certain discriminative classifiers are non-probabilistic: they directly estimate a function $f\colon X\to Y$ instead of $P(Y|X)$.) For clarity, we highlight $\boldsymbol{X}$ in bold to emphasize that it usually represents a vector of multiple attributes, i.e., $\boldsymbol{X}=\{X_1,X_2,\ldots,X_n\}$. However, this question does not require students to derivethe answer in vector/matrix notation.

In class we have observed an interesting relationship between a discriminative classifier (logistic regression) and a generative classifier (Gaussian naive Bayes): the form of $P(Y|\boldsymbol{X})$ derived from the assumptions of a specific class of Gaussian naive Bayes classifiers is precisely the form used by logistic regression. *The derivation can be found in the required reading: http://www.cs.cmu.edu/~tom/mlbook/NBayesLogReg.pdf*.We made the following assumptions for Gaussian naive Bayes classifiers to model $P(\boldsymbol{X},Y)=P(Y)P(\boldsymbol{X}|Y)$:

(1) Y is a boolean variable following a Bernouli distribution, with parameter $\pi=P(Y=1)$ and thus $P(Y=0)=1-\pi$.

(2) $\boldsymbol{X}=\{X_1,X_2,\ldots,X_n\}$, where each attribute X_i is a continuous random variable. For each X_i, $P(X_i|Y=k)$ is a Gaussian distribution $\mathcal{N}(\mu_{ik},\sigma_i)$. Note that σ_i is the standard deviation of the Gaussian distribution (and thus σ_i^2 is the variance), which does not depend on k.

(3) For all $i\neq j$, X_i and X_j are conditionally independent given Y. This is why this type of classifier is called "naive".

We say this is a specific class of Gaussian naive Bayes classifiers because we have made an assumption that the standard deviation σ_i of $P(X_i|Y=k)$ does not depend on the value k of Y. This is not a general assumption for Gaussian naive Bayes classifiers.

Let's make our Gaussian naive Bayes classifiers a little more general by removing the assumption that the standard deviation σ_i of $P(X_i|Y=k)$ does not depend on k. As a result, for each X_i, $P(X_i|Y=k)$ is Gaussian distribution $\mathcal{N}(\mu_{ik},\sigma_{ik})$, where $i=1,2,\ldots,n$ and $k=0,1$. Note that now the standard deviation σ_{ik} of $P(X_i|Y=k)$ depends on both the attribute index i and the value k of Y.

Question: is the new form of $P(Y|\boldsymbol{X})$ implied by this more general Gaussian naive Bayes classifier still the form used by logistic regression? Derive the new form of $P(Y|\boldsymbol{X})$ to prove your answer.

★ **SOLUTION:** **No**, the new $P(Y|\mathbf{X})$ implied by this more general Gaussian naive classifier is no longer the form used by logistic regression. Here is the derivation.

As shown in page 8 of [Mitchell: Naive Bayes and Logistic Regression], we have:

$$\begin{aligned}
P(Y=1|\mathbf{X}) &= \frac{P(Y=1)P(\mathbf{X}|Y=1)}{P(Y=1)P(\mathbf{X}|Y=1)+P(Y=0)P(\mathbf{X}|Y=0)} \\
&= \frac{1}{1+\frac{P(Y=0)P(\mathbf{X}|Y=0)}{P(Y=1)P(\mathbf{X}|Y=1)}} \\
&= \frac{1}{1+\exp(\ln\frac{P(Y=0)P(\mathbf{X}|Y=0)}{P(Y=1)P(\mathbf{X}|Y=1)})} \\
&= \frac{1}{1+\exp(\ln\frac{1-\pi}{\pi}+\ln\frac{P(\mathbf{X}|Y=0)}{P(\mathbf{X}|Y=1)})} \\
&= \frac{1}{1+\exp(\ln\frac{1-\pi}{\pi}+\sum_i\ln\frac{P(X_i|Y=0)}{P(X_i|Y=1)})}
\end{aligned}$$

Now the standard deviation σ_{ik} of $P(X_i|Y=k)$ depends on both the attribute index i and the value k of Y, so we have:

$$\begin{aligned}
\sum_i \ln\frac{P(X_i|Y=0)}{P(X_i|Y=1)} &= \sum_i \ln\frac{\frac{1}{\sqrt{2\pi\sigma_{i0}^2}}\exp(\frac{-(X_i-\mu_{i0})^2}{2\sigma_{i0}^2})}{\frac{1}{\sqrt{2\pi\sigma_{i1}^2}}\exp(\frac{-(X_i-\mu_{i1})^2}{2\sigma_{i1}^2})} \\
&= \sum_i \ln\frac{\sigma_{i1}}{\sigma_{i0}}+\sum_i(\frac{(X_i-\mu_{i1})^2}{2\sigma_{i1}^2}-\frac{(X_i-\mu_{i0})^2}{2\sigma_{i0}^2}) \\
&= \sum_i \ln\frac{\sigma_{i1}}{\sigma_{i0}}+\sum_i\frac{(\sigma_{i0}^2-\sigma_{i1}^2)X_i^2+2(\mu_{i0}\sigma_{i1}^2-\mu_{i1}\sigma_{i0}^2)X_i+\mu_{i1}^2\sigma_{i0}^2-\mu_{i0}^2\sigma_{i1}^2}{2\sigma_{i0}^2\sigma_{i1}^2} \\
&= \sum_i(\ln\frac{\sigma_{i1}}{\sigma_{i0}}+\frac{\mu_{i1}^2\sigma_{i0}^2-\mu_{i0}^2\sigma_{i1}^2}{2\sigma_{i0}^2\sigma_{i1}^2})+\sum_i\frac{(\mu_{i0}\sigma_{i1}^2-\mu_{i1}\sigma_{i0}^2)}{\sigma_{i0}^2\sigma_{i1}^2}X_i+\sum_i\frac{(\sigma_{i0}^2-\sigma_{i1}^2)}{2\sigma_{i0}^2\sigma_{i1}^2}X_i^2
\end{aligned}$$

As a result, we have $P(Y=1|\mathbf{X})$ as follows:

$$\begin{aligned}
P(Y=1|\mathbf{X}) &= \frac{1}{1+\exp(\ln\frac{1-\pi}{\pi}+\sum_i\ln\frac{P(X_i|Y=0)}{P(X_i|Y=1)})} \\
&= \frac{1}{1+\exp(w_0+\sum_i w_iX_i+\sum_i v_iX_i^2)}
\end{aligned}$$

where

$$\begin{aligned}
w_0 &= \ln\frac{1-\pi}{\pi}+\sum_i(\ln\frac{\sigma_{i1}}{\sigma_{i0}}+\frac{\mu_{i1}^2\sigma_{i0}^2-\mu_{i0}^2\sigma_{i1}^2}{2\sigma_{i0}^2\sigma_{i1}^2}) \\
w_i &= \frac{(\mu_{i0}\sigma_{i1}^2-\mu_{i1}\sigma_{i0}^2)}{\sigma_{i0}^2\sigma_{i1}^2} \\
v_i &= \frac{(\sigma_{i0}^2-\sigma_{i1}^2)}{2\sigma_{i0}^2\sigma_{i1}^2}
\end{aligned}$$

In general we do not have $\sigma_{i1}=\sigma_{i0}$ so the quadratic term $v_iX_i^2$ in $P(Y=1|\mathbf{X})$ will not disappear. Therefore the form of $P(Y=1|\mathbf{X})$ is no longer the form of logistic regression.

5. Programming (40 pts)

Solution:
Please check the four attachments.